METHYLERGOMETRINE

Never in hypertension/pre-eclampsia

Store in fridge

CHEMICAL NAME (8β)-N-[(2S)-1-hydroxybutan-2-yl]-6-methyl-9,10-didehydroergoline-8-carboxamide (maleate)

CLASS Ergot alkaloid uterotonic (methylergonovine)

TRADE NAMES Methergine · Methergin · Ergometrine (related) · Syntometrine (ergometrine + oxytocin)

THERAPEUTIC USES / INDICATIONS

- Postpartum haemorrhage due to uterine atony (second-line after oxytocin).
- Subinvolution of the uterus.
- Active management of third stage (combined with oxytocin in some regions).

ADMINISTRATION

- **IM** (preferred) or **IV slowly over ≥60 s** (IV only in emergencies — hypertension, stroke risk).
- **PO** for subinvolution.
- Never before delivery of the baby (and placenta in some protocols).
- Store at 2–8 °C, protect from light.

DOSAGES

- **PPH:** 200 mcg (0.2 mg) IM, repeat every 2–4 h if needed.
- **PO subinvolution:** 200 mcg 3–4 times daily for up to 7 days.

PHARMACOKINETICS

- **Onset:** IV 1 min; IM 2–5 min; PO 5–10 min. **Duration:** 3 h (sustained tetanic contraction).
- Metabolised in liver (CYP3A4).
- Excreted in bile/urine. **t½:** 3 h.

PHARMACODYNAMICS

- Stimulates **α-adrenergic and serotonin (5-HT2)** receptors on uterine smooth muscle → sustained, tetanic contraction → compresses bleeding vessels.
- Also causes vasoconstriction → ↑ BP.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting.
- Headache, dizziness.
- Abdominal cramps.

ADVERSE EFFECTS (SERIOUS)

- Severe hypertension, stroke, seizures.
- Coronary vasospasm, MI.
- Peripheral ischaemia (ergotism) with CYP3A4 inhibitors.

CONTRAINDICATIONS & CAUTIONS

- Hypertension, pre-eclampsia/eclampsia.
- Cardiac disease, peripheral vascular disease.
- Induction of labour; before delivery of the baby.
- Severe hepatic/renal impairment; sepsis.

KEY INTERACTIONS

- **Strong CYP3A4 inhibitors (macrolides, azoles, HIV PIs):** ergotism; avoid.
- **Triptans, other vasoconstrictors:** additive vasoconstriction.
- **Halothane:** ↓ uterotonic effect.

NURSING MANAGEMENT

- Check BP before giving; withhold if hypertensive or pre-eclamptic.
- Monitor BP, HR, uterine tone and blood loss after dose.
- Report headache, chest pain, visual disturbance.
- Store correctly; check expiry.

PATIENT EDUCATION

- Report severe headache, chest pain, numbness or coldness of hands/feet.
- Cramping indicates the uterus is contracting.

KEY POINTS

- Ergot uterotonic — sustained contraction.
- Never in hypertension or pre-eclampsia.
- PPH: 200 mcg IM.
- Avoid with macrolides/azoles.

SOURCES CHECKED **[670]** Methergine, oral (US label) **[671]** Methylergonovine Maleate, IM, IV (US label)

Full references p. 432–457